TEXORA · LECTURE SHEET

Picking

Inserting one pick across the shed: how cam, stick and strap throw the shuttle from box to box, why crank-circle timing is unforgiving, and what happens when the shuttle is weak, late, trapped or unchecked.

TE-2103 · WEAVING TECHNOLOGY · TOPIC 7 · LEVEL 2 TERM 1

1. What picking is

Picking is the insertion of a single length of weft — one **pick** — across the width of the warp shed, from one side of the loom to the other. On a conventional shuttle loom the pick is not blown, projected or carried by a gripper: it is carried inside a boat-shaped **shuttle** that flies freely across the loom, laying the weft behind it from a **pirn** wound inside the shuttle body. It is the second primary motion: shedding opens the warp, picking puts the weft in, beat-up pushes it home.

One pick is inserted per crankshaft revolution. At 200 picks per minute the shuttle is thrown 200 times a minute, alternately left-to-right and right-to-left, yet it flies for only about one third of each revolution, resting in a box the rest of the time. A heavy shuttle must be accelerated from rest, flown across the full reed width and stopped again inside that short window — and its force, timing and checking all follow from that fact.

2. The picking mechanism of a shuttle loom

The mechanism converts continuous rotation into one sharp blow per revolution, alternately at the two ends of the sley. The chain of parts is the same on almost every conventional loom.

2.1 The parts, in order of action

- **Bottom shaft.** The lower shaft, driven from the crankshaft, carrying the picking cams so that a cam acts once per pick at each side.
- **Picking cam and nose.** A cam on the bottom shaft whose profile is mostly a dwell, with one sharply rising projection — the **nose**. As the nose passes the picking stick it displaces it suddenly. A longer, gentler nose gives a smoother start; a short, steep nose gives a harder blow.
- **Picking stick.** A vertical wooden or laminated stick pivoted at its lower end. The cam nose drives its lower part; its top swings rapidly toward the loom centre, carrying the strap with it.

- **Picking strap and lug (picker).** A leather strap connects the stick top to the **lug** — a block sliding in the shuttle box against the shuttle end. When the stick swings, the strap pulls the lug forward and it strikes the shuttle out of the box.
- **Shuttle box.** Box at each sley end housing the shuttle at rest (base: race; front: reed). It must hold the shuttle in line with the race — neither rattling nor braking it.
- **Shuttle and pirn.** A hollowed wooden or composite body, steel-tipped at both ends, carrying a **pirn** — a small tapered bobbin of weft on a spring spindle inside the cavity. Weft unwinds through a porcelain eye in the shuttle wall during flight. A typical cotton-loom shuttle weighs about 350–450 g complete.
- **Check strap and swell.** At the receiving end a leather **check strap** buffers the shuttle end, and the **swell** — a spring-loaded projection in the box back — presses on the side of the arriving shuttle. Together their friction absorbs the shuttle's kinetic energy within the length of the box.

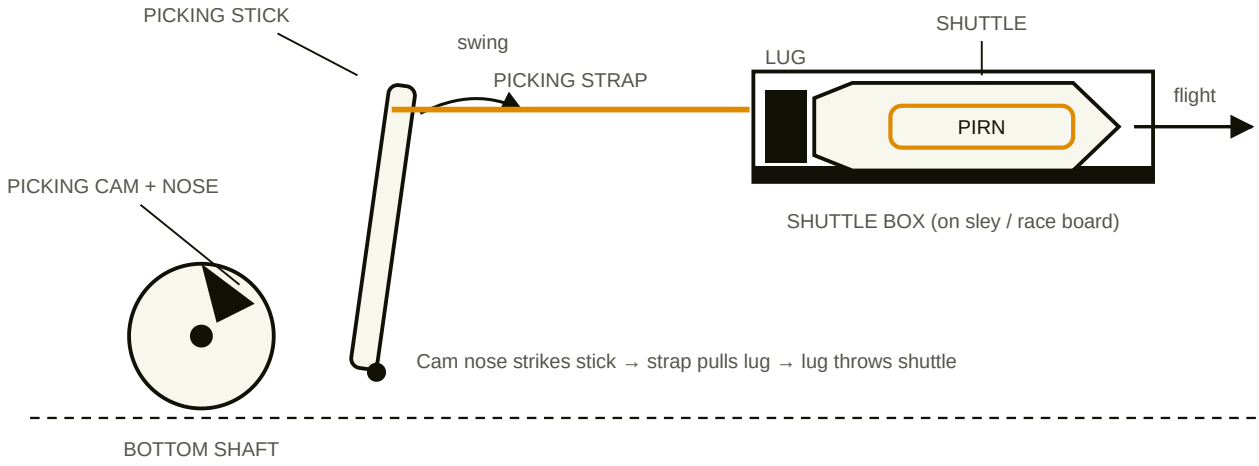


Figure 1 — Picking mechanism, side view (one side of the loom). The picking cam on the bottom shaft drives the picking stick; the strap pulls the lug through the shuttle box and the lug projects the shuttle, with its pirn, across the shed.

2.2 The picking sequence, step by step

1. The sley moves back after beat-up; the shed opens and the shuttle rests in its box, held by the swell.
2. The bottom shaft turns and the nose of the picking cam reaches the picking stick.
3. The nose displaces the stick; the stick pivots and its top swings sharply toward the loom centre.
4. The strap pulls the lug along the box; the lug strikes the steel tip of the shuttle.
5. The shuttle accelerates along the box, leaves it onto the race and enters the open shed at full speed, weft unwinding from the pirn.
6. It flies across, guided below by the race and behind by the reed, laying the pick behind it.
7. At the far side it enters the opposite box, meets swell and check strap, and comes to rest. The shed changes, trapping the pick, and the sley comes forward for beat-up.

3. Overpick and underpick mechanisms

3.1 Overpick mechanism

In the **overpick** system the picking stick is mounted **above** the shuttle box and the cam works it indirectly, through levers and straps. It is the classic arrangement on older and medium-speed looms: accessible and easy to inspect, but the longer, jointed drive loses motion as straps stretch and is harder to keep identical at the two sides.

3.2 Underpick mechanism

In the **underpick** system the picking stick is mounted **below** the shuttle box, pivoted low on the sley sword, and the cam acts directly on the stick, as in Figure 1. The drive is short, with fewer joints, so timing is crisper and higher speeds are possible. The parts are less accessible, crowded low in the loom, and the cam nose and stick face wear at their direct contact and need careful lubrication.

FEATURE	OVERPICK	UNDERPICK
Position of picking stick	Above the shuttle box	Below the shuttle box
Cam action	Indirect — cam works the stick through levers / straps	Direct — cam nose acts on the stick (or its roller)
Length of drive chain	Longer, more pivots and straps	Shorter, fewer parts
Lost motion / timing sharpness	More lost motion as straps stretch	Less lost motion; sharper timing
Typical use	Older and medium-speed looms	Higher-speed conventional looms

4. Picking timing on the crank circle

Loom timing is read on the **crank circle**: one circle = one crank revolution, one pick, one beat-up. **0° (360°) is front dead centre — beat-up**, the reed closest to the fell; angles increase as the sley retreats, then returns.

Picking must satisfy three conditions together:

- The **shed must be open** enough (typically 25–35 mm at the shuttle path) for the shuttle to pass without striking warp.
- The **reed must be near back centre**, with the race clear and a straight, supported shuttle path.
- The shuttle must have **time to cross and settle** before the shed closes and the sley runs forward.

Picking therefore **begins at about 100–110°** after beat-up. The shuttle flies for roughly the next 100–130° and should be boxed and settled by about 220–240°. Earlier risks a shed not yet open; later shortens the settling margin and invites a trapped shuttle.

Timing is set by rotating the cam on the bottom shaft, checked by turning the loom by hand to where the lug just moves the shuttle and reading the crank angle, at both sides — the *crank angle* must match.

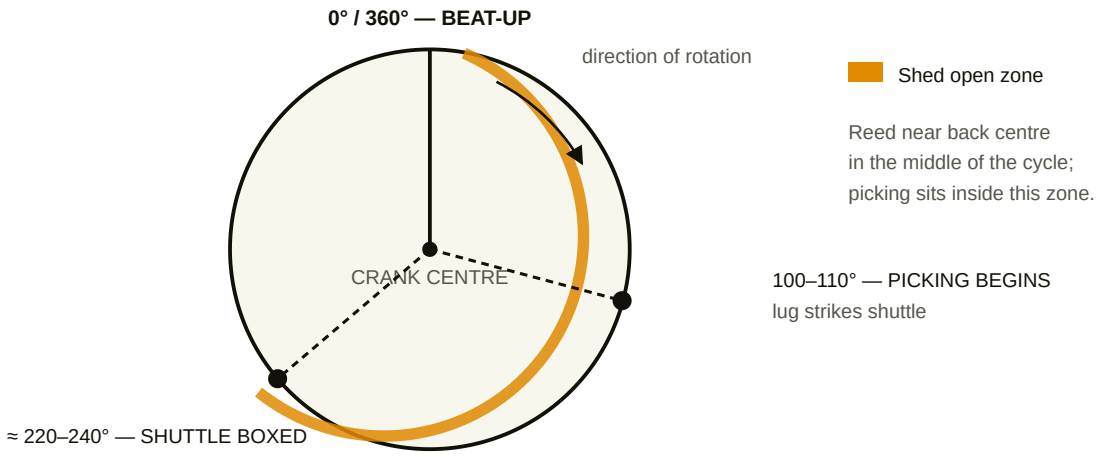


Figure 2 — Picking timing on the crank circle. 0° is beat-up at front dead centre. Picking begins at about 100–110°, inside the shed-open zone with the reed near back centre, and the shuttle must be boxed by about 220–240°.

5. Shuttle flight and checking

5.1 Flight

Once it leaves the box the shuttle is a free projectile guided by contact: its underside slides on the **shuttle race** and its back rides lightly against the **reed**. Flight must be straight and level; a shuttle leaving at an angle, or struck off-centre by a worn lug, will hit the reed or warp, ride up, or dive into the lower shed sheet. Speed is highest just after leaving the box and falls steadily to friction and weft drag, so the *mean* speed calculated below is below the exit peak, which may be 25–30 % higher.

Weft tension in flight comes only from unwinding the pirn (firm base, conical chase); a badly wound pirn gives tight picks, selvedge loops, or a mid-flight break — a **broken pick**.

5.2 Checking

Checking is stopping the shuttle in the receiving box: all its KE must be lost within roughly one shuttle length, by **swell** friction and the **check strap** buffer. It must be firm but not violent:

- **Too little checking** (worn swell, weak spring, slack strap) lets the shuttle rebound into the shed or protrude into the reed's path.

- **Too much checking** (swell too hard, box too tight) brakes the shuttle short of home, wastes power, and heats and wears the shuttle.

Correct checking is judged running: the shuttle should slide home and stay, with a dull sound — not a crack.

WORKED EXAMPLE 1 — MEAN SHUTTLE VELOCITY

A loom runs at 200 picks/min and the reed width is 1.6 m. Shuttle flight occupies 120° of crank rotation. Find the flight time and the mean shuttle velocity.

Step 1 — Time for one crank revolution: 60 s ÷ 200 revolutions = **0.30 s per revolution**.

Step 2 — Flight time: flight is 120° of 360°, so time = (120 ÷ 360) × 0.30 s = (1/3) × 0.30 = **0.10 s**.

Step 3 — Mean velocity: v = distance ÷ time = 1.6 m ÷ 0.10 s = **16 m/s**.

About 58 km/h — to be stopped inside one box length at the far side.

WORKED EXAMPLE 2 — SHUTTLE KINETIC ENERGY

For the same loom, the shuttle complete with pirn has a mass of 0.4 kg and a mean flight speed of 16 m/s (Worked example 1). Estimate the kinetic energy that checking must absorb, using the mean speed.

Step 1 — Formula: KE = ½ × m × v^2.

Step 2 — Substitute: KE = ½ × 0.4 kg × $(16\ \text{m/s})^2$ = 0.2 × 256 = **51.2 J**.

Step 3 — Meaning: about 51 J must be turned into heat and sound by swell and check strap on every pick — why picking and checking dominate loom power and wear.

6. Picking power

Picking is the largest single consumer of power on a conventional loom: every pick must supply the shuttle's kinetic energy plus friction in drive, flight and checking. Since KE rises with the *square* of speed, double the flight speed means four times the energy per pick.

$$\text{KE} = \tfrac{1}{2}\, m\, v^2 \quad \cdot \quad v \approx \text{reed width} \div \text{flight time} \quad \cdot \quad \text{flight time} = (\text{flight angle} \div 360°) \times (60 \div \text{picks per minute})$$

Hence wider looms run slower, and heavier shuttles cost energy on every pick. A slack strap or glazed lug wastes the blow; strengthening the pick to compensate raises power, wear and noise together. Shuttleless looms exist largely to escape this cost.

7. Picking faults and their consequences

FAULT	WHAT HAPPENS	TYPICAL CAUSES
Shuttle trap (shuttle trapped / smash precursor)	Shuttle fails to clear the shed before it closes; caught between warp sheets.	Late or weak pick; shed closing early; shuttle slowed by a tight box, damaged race or dragging weft.
Smash	Trapped shuttle driven forward by the reed: many warp ends break at once. The warp protector should stop the loom first.	Undetected trap; protector motion faulty.
Shuttle fly-out	The shuttle leaves the loom altogether, flying off the race — a serious safety hazard.	Angled entry; bent tip; loose reed; misaligned box; race obstruction.
Shuttle rebound	The shuttle bounces back out of the receiving box into the shed or into the reed's path.	Checking too weak (worn swell, slack check strap) combined with an over-strong pick.
Weak pick	The shuttle does not reach the far box, or arrives too slowly and stops short.	Worn cam nose; slack strap; worn lug; tight box.
Late pick	The shuttle crosses correctly but too late in the cycle, narrowing the margin before shed closing.	Cam slipped or set late; lost motion in the drive.

Picking faults are rarely cured by force. Diagnose in order: timing, then alignment (box, race and reed in one line), then condition (cam, stick, strap, lug, swell, check strap), and only last the strength of the pick.

Glossary

Picking — inserting one pick across the shed. **Shuttle** — the free-flying weft carrier. **Pirn** — the tapered weft bobbin inside the shuttle. **Picking cam** — cam on the bottom shaft whose nose drives the stick once per pick. **Picking stick** — pivoted stick operating the strap. **Lug / picker** — sliding block in the box that strikes the shuttle. **Swell** — spring projection pressing on the arriving shuttle. **Check strap** — leather buffer finally stopping the shuttle. **Crank circle** — 360° diagram of one revolution, 0° at beat-up. **Smash** — mass warp breakage when a trapped shuttle is driven into the warp. **Overpick / underpick** — stick above / below the box.

Key points

• One pick per crank revolution; the shuttle flies freely with its pirn.
• Drive chain: bottom shaft → cam → stick → strap → lug → shuttle.

• The swell and check strap check the shuttle; checking must be firm, without rebound.
• Picking begins at about 100–110° on the crank circle — shed open, reed near back centre — and the shuttle should be boxed by about 220–240°.
• Mean flight speed is typically 14–18 m/s; shuttle KE (½mv²) dominates power and checking wear.
• Trap, rebound, fly-out, weak and late picks are diagnosed in the order: timing, alignment, condition — not by simply strengthening the pick.

Practice

1. At 180 ppm, reed width 1.4 m, flight 120°: calculate flight time and mean shuttle velocity.
2. A 0.45 kg shuttle flies at 15 m/s. Calculate its kinetic energy, showing every step.
3. Draw and label the crank circle: mark 0°, picking start, boxed position and shed-open zone, and state why picking cannot start at 20°.
4. List the picking parts in order from bottom shaft to shuttle, and state the jobs of swell and check strap.
5. A loom suffers repeated shuttle rebound at one side only. Give four checks, in diagnostic order.
6. Using KE = ½mv², explain why a wide loom runs at fewer ppm than a narrow loom with the same shuttle.

Answer key

1. 60 ÷ 180 = 0.333 s/rev; flight = (120 ÷ 360) × 0.333 = 0.111 s; v = 1.4 ÷ 0.111 ≈ **12.6 m/s**.
2. KE = ½ × 0.45 × 15² = 0.225 × 225 = **50.6 J**, absorbed in checking every pick.
3. 0° at beat-up (top); picking at 100–110°; boxed at about 220–240°; shed-open zone shaded around back centre. At 20° the sley has barely left the fell and the shed is not yet open, so the shuttle would strike warp and reed.
4. Bottom shaft → cam and nose → stick → strap → lug → shuttle (with pirn). The swell brakes the arriving shuttle by side friction; the check strap is the final buffer stopping it without rebound.
5. (1) Timing early/strong; (2) box alignment; (3) swell/check strap worn or slack; (4) over-strong pick from cam, strap or lug.
6. At the same ppm flight time is equal, so width needs proportionally higher speed; KE rises with v², hence the wide loom runs slower.